

Research and Practice of Optical Network Technology for High Reliability Interconnection of Large-scale Data Centers

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Abstract: We have demonstrated a scalable optical network architecture integrating 800 Gbit/s or beyond coherent transmission, optical circuit switching, and telecom-grade ultra-fast rerouting technologies to enable highly reliable, low-latency and large-scale inter- and intra-data center networks, validated in both lab and field. © 2026 The Author(s)

1. Introduction

The explosive growth of large-scale artificial intelligence (AI) training and high-performance distributed inference is reshaping data center architectures and elevating optical networking from a transport substrate to a production-critical infrastructure [1,2]. As model sizes and cluster scales increase, the number of accelerators (XPU) in AI data centers (AIDCs) grows rapidly, intensifying cross-device communication and synchronization. In large-scale synchronous training, where thousands of devices iteratively exchange gradients to maintain model consistency, system throughput becomes tightly coupled to communication latency and failure events. Under such workloads, even short network disruptions can stall synchronization and strand accelerators, translating reliability directly into measurable computational and economic cost. Recent industry analyses indicate that at an operational cost of approximately \$2.5 per GPU per hour, a 10% downtime attributable to network and optical failures may result in daily losses of roughly \$96,000 for a 16K-GPU cluster and \$3 million for a 512K-GPU-scale deployment [3].

Standardization and industry visions such as ION-2030 emphasize the evolution of optical networks toward sustainable, intelligent, and resilient connectivity to support emerging AI services. In this context, the AI data center (AIDC) becomes the physical core of AI computing infrastructure, where optical interconnection forms the fundamental fabric enabling large-scale distributed workloads [4,5]. Inside a server node, GPUs communicate over electrical interconnects (e.g., NVLink/PCIe), and GPU-to-network interface card (NIC) transfers are realized via high-speed system buses, often with GPUDirect RDMA support. Optical interconnects begin beyond the NIC. In data center intra-connection network (DCN), server-to-switch and switch-to-switch links primarily rely on short-reach optical modules forming high-bandwidth fabrics; while in data center inter-connection network (DCI), long-reach coherent optics and WDM transport systems interconnect geographically separated data centers over fiber.

In this paper, we analyze the impact of optical link failures on AI workloads in large-scale data center. For DCI, we investigate the interaction between fiber disruptions and collective synchronization, and validate telecom-grade ultra-fast rerouting technology (sub-50 ms optical-layer protection) on a field-deployed four-site distributed training network, demonstrating effective confinement of disruptions within synchronization tolerance. For DCN, we implement and verify an optical circuit switching (OCS)-based architecture on the field-deployed network to enhance fabric-level resilience and reduce optical module exposure. The results show that recovery time and optical architecture design as critical factors for high reliability interconnection of large-scale data centers.

2. Failure Scenarios in Cross-NIC/DC Synchronous Training Communication

Figure 1 illustrates representative communication states between two network interface cards (NICs) located in different data centers and interconnected through an optical fiber link. For clarity and to focus on end-to-end synchronous training behavior, intermediate components such as optical modules and switching nodes are abstracted in the diagram. The communication model assumes ordered packet transmission with sequence numbers and loss detection via negative acknowledgments (NAKs), consistent with RDMA-based reliable transports widely adopted in large-scale synchronous AI training clusters. In synchronous distributed training, each iteration consists of a compute phase followed by a collective gradient synchronization phase. If an optical fiber disruption occurs during the compute phase and is restored before the next synchronization window begins, the training process remains algorithmically unaffected. In this case, the interruption is effectively masked by computation and does not propagate to collective communication. Therefore, the impact of optical-layer failure is fundamentally determined by its temporal alignment with the collective synchronization window, rather than by the raw outage duration alone.

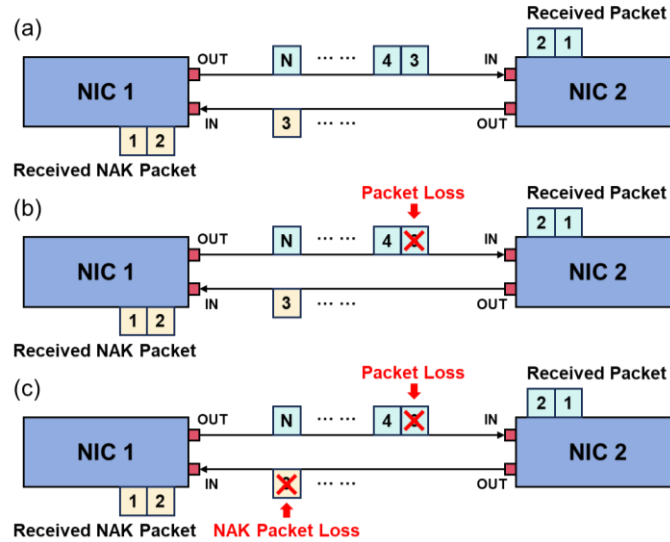


Fig. 1: (a) Simplified schematic of cross-DC optical interconnection and failure scenarios during synchronous training: (a) normal synchronization; (b) forward packet loss with NAK-based recovery; (c) bidirectional packet and NAK loss with timeout-driven recovery. NIC: network interface card.

(a) Normal Operation: Under normal conditions, in Fig. 1(a), data packets are delivered in order and control feedback is returned successfully. The sender advances its transmission window without retransmission, and the synchronization round completes deterministically within the expected communication interval.

(b) Forward Data Packet Loss with NAK Delivered (RTT-Bounded Recovery): In Fig. 1(b), a forward data packet (packet 3) is lost due to a transient optical-layer impairment, while the reverse feedback path remains operational. The receiver detects the sequence gap and issues a NAK corresponding to the first missing packet. Upon receiving the NAK, the transmitter initiates loss recovery. This is the so-called Go-Back-N mechanism in RDMA-based reliable transmission: retransmission starts from the first lost packet and include subsequent packets within the current transmission window. Consequently, at least one round-trip time (RTT) of data transmission is effectively repeated. Although this introduces additional latency and bandwidth overhead, recovery remains bounded by the RTT. Transport-layer state consistency is preserved, and the failure does not escalate beyond localized retransmission.

(c) Data Packet and NAK Loss (Timeout-Dominated Recovery): In Fig. 1(c), both the forward data packet and the corresponding NAK are lost due to a bidirectional optical impairment. From the transmitter's perspective, no loss indication is received. The transmission window appears valid until the transport-layer timeout expires. As a result, retransmission can only be triggered by timeout rather than by immediate NAK feedback. In practical AI communication stacks (e.g., HCCL-based implementations), the timeout threshold is typically configured on the order of seconds, commonly around 2 seconds depending on communication parameters. During this timeout interval, the transmitter may continue injecting data within the allowed transmission window without awareness of the loss condition. While the actual ineffective transmission volume depends on window size and flow-control constraints, this illustrates the significant amplification potential of timeout-driven recovery compared to RTT-scale NAK-triggered retransmission. Specially, when the final packet of a synchronization round and its corresponding NAK are both lost, the transmitter assumes completion of transmission, while the receiver cannot finalize the collective operation. Since no subsequent packets are available to trigger additional feedback, timeout-driven recovery becomes unavoidable. This creates temporary state ambiguity at the synchronization boundary and prolongs barrier completion.

(d) Extended Link Outage Exceeding Timeout (Connection-Level Problem): If the optical link experiences a sustained outage exceeding the transport timeout threshold, neither forward data nor reverse feedback can be exchanged within the allowable window. Once the timeout expires, the transport may escalate to connection-level error handling, such as queue pair reset or connection teardown, depending on implementation. Unlike cases (b) and (c), recovery is no longer confined to window retransmission. Instead, endpoint state may be invalidated, requiring reinitialization. In large-scale synchronous clusters, this may stall collective operations and strand a substantial number of GPUs at a synchronization barrier.

3. Field-deployed Network Practice

In DCI scenario, a full-mesh wavelength switched optical network () is established in Beijing, consisting of 4 reconfigurable optical add-drop multiplexer sites (ROADM, site A/B/C/D) and 3 optical line amplifier (OLA, site

E/F/G) sites as Fig. 2(a) shows. The intelligent computing AI servers are deployed in site D, with each server configured with 8 neural network processing units (NPUs), totaling 32 NPUs. These NPUs were divided into two groups, with two servers in each group, simulating two DC sites (DC1&DC2). DC1 and DC2 are communicated using an 800G 16-QAM optical transport unit (OTU) with a center frequency of 190.75 GHz.

The original signal route is directly from Site D to Site C. At first, a fiber break was simulated between Site D and Site C, and the signal was rerouted through Site D→Site B→Site C. Next, a fiber cut was introduced between Site D and Site G, causing the signal to reroute via Site D→Site A→Site C. To emulate rapid optical power degradation, EDFA pump lasers at Sites F and G were intermittently shut down. In all scenarios, service restoration was completed within 50 ms [6]. The sub-50 ms optical-layer protection exhibited negligible impact on computational performance, with training throughput remaining stable and TFLOPS decreasing only marginally during protection events. No communication interruption was observed throughout the tests.

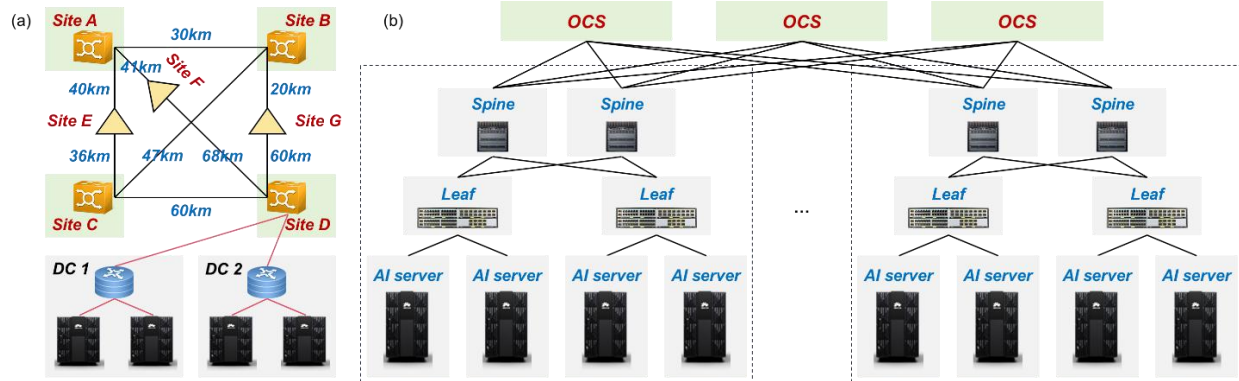


Fig. 2: (a) DCI: 4-sites field trial, (b) DCN: OCS-based field trial.

In the DCN scenario, an optical circuit switching (OCS)-enabled fabric architecture was deployed and validated on a field-deployed network. AI servers were interconnected through a dynamically reconfigurable optical-electrical coordinated network, where OCS was introduced to provide flexible optical path provisioning and reduce reliance on static point-to-point optical module connections.

To evaluate resilience under large-scale AI traffic conditions, dynamic optical path reconfiguration and traffic switching operations were performed during ongoing AI workloads. The OCS-based architecture enabled rapid optical path adjustment and maintained service continuity during reconfiguration events. The results indicate that GPU-to-GPU point-to-point latency is reduced by approximately 14%, from 8.21 μ s to 7.18 μ s, owing to simplified forwarding paths and fewer intermediate electrical switching stages. By transitioning from a three-stage electrical switching structure to a two-stage optical-electrical coordinated architecture, overall interconnection power consumption decreases by approximately 19%, accompanied by an estimated 17% improvement in system-level reliability due to reduced optical module exposure. No service interruption was observed in any test scenario, and computational performance remained stable during dynamic switching operations.

4. Conclusions

We have investigated optical-layer reliability for large-scale AI data center interconnection across both DCI and DCN domains. The results show that disruption impact in distributed training is governed by its alignment with synchronization windows rather than outage duration alone. Field validation on a four-site network confirms that telecom-grade ultra-fast rerouting technology (sub-50 ms optical-layer protection) effectively constrains fiber impairments within synchronization tolerance, and an OCS-enabled DCN architecture enhances fabric resilience and reduces optical module exposure. We show a feasibility of next-generation optical network technologies for achieving high-reliability interconnection in large-scale data centers.

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